$$\vec{F}_{12} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}(-\hat{r}_{12}) = -\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}(\hat{r}_{12})$$

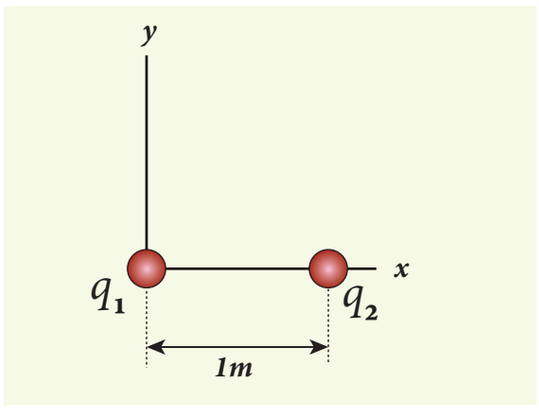
(or) $\vec{F}_{12} = -\vec{F}_{21}$

Therefore, the electrostatic force obeys Newton's third law.

(viii) The expression for Coulomb force is true only for point charges. But the point charge is an ideal concept. However we can apply Coulomb's law for two charged objects whose sizes are very much smaller than the distance between them. In fact, Coulomb discovered his law by considering the charged spheres in the torsion balance as point charges. The distance between the two charged spheres is much greater than the radii of the spheres.

EXAMPLE 1.2

Consider two point charges q_1 and q_2 at rest as shown in the figure.



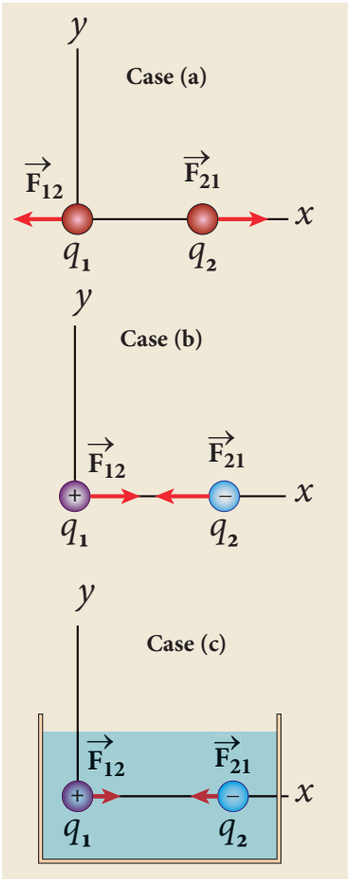
They are separated by a distance of 1m. Calculate the force experienced by the two charges for the following cases:

(a) $q_1 = +2\mu C$ and $q_2 = +3\mu C$

(b) $q_1 = +2\mu C$ and $q_2 = -3\mu C$

(c) $q_1 = +2\mu C$ and $q_2 = -3\mu C$ kept in water ($\varepsilon_r = 80$)

Solution



(a) $q_1 = +2\ \mu C$, $q_2 = +3\ \mu C$, and $r = 1m$. Both are positive charges. so the force will be repulsive

Force experienced by the charge q_2 due to q_1 is given by

$$\vec{F}_{21} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{r}_{12}$$

Here $\hat{r}_{12}$ is the unit vector from q_1 to q_2. Since q_2 is located on the right of q_1, we have

$\hat{r}_{12} = \hat{i}$, so that

$$\vec{F}_{21} = \frac{9\times10^9 \times 2\times10^{-6} \times 3\times10^{-6}}{1^2}\hat{i} \quad \left[\frac{1}{4\pi\varepsilon_0} = 9\times10^9\right]$$

$$= 54\times10^{-3} N\hat{i}$$

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